

LAB	PRACTICALS
13	Implementation of Aggregate Functions in SQL (Part – 1) Create STUDENT_MARKS table with SID INT, SName VARCHAR(25) and Marks INT. Insert the records into the Student_Marks table and perform the following queries: Part - A - Find the total number of students. <code>SELECT COUNT(*) AS Total_Students FROM STUDENT_MARKS;</code> - Find the total marks scored by all students. <code>SELECT SUM(Marks) AS Total_Marks FROM STUDENT_MARKS;</code> - Find the average marks of all students. <code>SELECT AVG(Marks) AS Average_Marks FROM STUDENT_MARKS;</code> - Find the minimum marks scored by any student. <code>SELECT MIN(Marks) AS Minimum_Marks FROM STUDENT_MARKS;</code> - Find the maximum marks scored by any student <code>SELECT MAX(Marks) AS Maximum_Marks FROM STUDENT_MARKS;</code> Part – B Aggregate Functions with GROUP BY and GROUP BY Having Clause - Display the Highest, Lowest, Total, and Average salary of all employees & label the columns Maximum, Minimum, Total_Sal and Average_Sal, respectively. <code>SELECT MAX(Salary) AS Maximum, MIN(Salary) AS Minimum, SUM(Salary) AS Total_Sal, AVG(Salary) AS Average_Sal FROM Employee;</code> - Find total number of employees of EMPLOYEE table. <code>SELECT COUNT(*) AS Total_Employees FROM Employee;</code> - Retrieve maximum salary from IT department. <code>SELECT MAX(Salary) AS Max_IT_Salary FROM Employee WHERE Department = 'IT';</code> - Count total number of cities of employee without duplication. <code>SELECT COUNT(DISTINCT City) AS Total_Cities FROM Employee;</code> - Display city with the total number of employees belonging to each city. <code>SELECT City, COUNT(*) AS Total_Employees FROM Employee GROUP BY City;</code> - Display city having more than one employee. <code>SELECT City, COUNT(*) AS Total_Employees</code>

	FROM Employee GROUP BY City HAVING COUNT(*) > 1; 7. Give total salary of each department of EMPLOYEE table. SELECT Department, SUM(Salary) AS Total_Salary FROM Employee GROUP BY Department; 8. Give average salary of each department of EMPLOYEE table without displaying the respective department name. SELECT AVG(Salary) AS Average_Salary FROM Employee GROUP BY Department; 9. Give minimum salary of employee who belongs to Ahmedabad. SELECT MIN(Salary) AS Min_Salary_Ahmedabad FROM Employee WHERE City = 'Ahmedabad'; 10. List the departments having total salaries more than 50000 and located in city Rajkot. SELECT Department, SUM(Salary) AS Total_Salary FROM Employee WHERE City = 'Rajkot' GROUP BY Department HAVING SUM(Salary) > 50000;
14	Implementation of Aggregate Functions in SQL (Part – 2) Create the following table using Design Mode, insert the given data and perform the following queries: Part - A 1. Display the Highest, Lowest SPI of all students & label the columns Maximum, Minimum respectively. SELECT MAX(SPI) AS Maximum, MIN(SPI) AS Minimum FROM STUDENT_DATA 2. Find total number of students of student table. SELECT COUNT(SID) AS TOTAL_STUDENTS FROM Student_Data 3. Retrieve maximum SPI from BCA department. SELECT MAX(SPI) AS MAX_SPI FROM Student_Data WHERE Department = 'BCA' 4. Count total number of cities of students without duplication. SELECT COUNT(DISTINCT CITY) AS TOTAL_CITIES FROM Student_Data 5. Display city with the total number of students belonging to each city.

```
SELECT CITY,COUNT(SID) AS TOTAL_STUDENTS
FROM Student_Data
GROUP BY CITY
```

Part – B

6. Display city having more than one student.

```
SELECT CITY,COUNT(SID) AS STUDENT_COUNT
FROM Student_Data
GROUP BY CITY
HAVING COUNT(SID)>1
```

7. Give total SPI of each department of student table.

```
SELECT DEPARTMENT,SUM(SPI) AS TOTAL_SPI
FROM Student_Data
GROUP BY Department
```

8. Give average SPI of each department of student table without displaying the respective department name.

```
SELECT AVG(SPI) AS AVERAGE_SPI
FROM Student_Data
GROUP BY Department
```

9. Give minimum SPI of student who belongs to Ahmedabad.

```
SELECT MIN(SPI) AS MIN_SPI
FROM Student_Data
WHERE CITY = 'AHMEDABAD'
```

10. List the departments having total SPI more than 15

```
SELECT DEPARTMENT,SUM(SPI) AS TOTAL_SPI
FROM Student_Data
GROUP BY Department
HAVING SUM(SPI)>15
```

Part – C

11. Display department having more than one student.

```
SELECT DEPARTMENT,COUNT(SID) AS STUDENT_COUNT
FROM Student_Data
GROUP BY Department
HAVING COUNT(SID)>1
```

12. Find total number of students from MCA department.

```
SELECT COUNT(SID) AS STUDENT_COUNT
FROM Student_Data
WHERE Department = 'MCA'
```

13. Retrieve maximum SPI from BBA department.

```
SELECT MAX(SPI) AS MAX_SPI
FROM Student_Data
WHERE Department = 'BBA'
```

14. Count total number of department of students table without duplication.

```
SELECT COUNT(DISTINCT DEPARTMENT) AS DEPT_COUNT
```

	FROM Student_Data 15. List the departments having total number of students more than one and belonging to <i>Rajkot</i> city. <pre>SELECT Department, COUNT(*) AS Total_Employees FROM Employee WHERE City = 'Rajkot' GROUP BY Department HAVING COUNT(*) > 1;</pre>
15	Practice the basic SQL commands, various built-in functions (Math, String, Date), Aggregate functions and Group by Having clause. Create the following table using Design Mode, insert the given data and perform the following queries: - Display all the employees whose name starts with “m” and 4th character is “h”. <pre>SELECT * FROM Employees WHERE EName LIKE 'm__h%';</pre> - Find the value of 3 raised to 5. Label the column as output. <pre>SELECT POWER(3,5) AS output;</pre> - Write a query to subtract 20 days from the current date. <pre>SELECT DATEADD(DAY, -20, GETDATE()) AS ResultDate;</pre> - Write a query to display name of employees whose name starts with “j” and contains “n” in their name. <pre>SELECT EName FROM Employees WHERE EName LIKE 'j%n%';</pre> - Display 2nd to 9th character of the given string “SQL Programming”. <pre>SELECT SUBSTRING('SQL Programming', 2, 8) AS Result;</pre> - Display name of the employees whose city name ends with “ney” & contains six characters. <pre>SELECT EName FROM Employees WHERE City LIKE '____ney';</pre> - Write a query to convert value 15 to string. <pre>SELECT CAST(15 AS VARCHAR(10)) AS Result;</pre> - Add department column with varchar(20) to Employees table. <pre>ALTER TABLE Employees ADD Department VARCHAR(20);</pre> - Set the value of department to Marketing who belongs to London city. <pre>UPDATE Employees SET Department = 'Marketing'</pre>

```
WHERE City = 'London';
```

10. Display all the employees whose name ends with either "n" or "y".

```
SELECT *FROM Employees  
WHERE EName LIKE '%n' OR EName LIKE '%y';
```

11. Find smallest integer value that is greater than or equal to 63.1, 63.8 and -63.2.

```
SELECT CEILING(63.1) AS Val1,  
CEILING(63.8) AS Val2,  
CEILING(-63.2) AS Val3;
```

12. Display all employees whose joining date is not available.

```
SELECT *  
FROM Employees  
WHERE JoiningDate IS NULL;
```

13. Display name of the employees in capital letters and city in small letters.

```
SELECT UPPER(EName) AS Name,  
LOWER(City) AS City  
FROM Employees;
```

14. Change the data type of Ename from varchar(30) to char(30).

```
ALTER TABLE Employees  
ALTER COLUMN EName CHAR(30);
```

15. Display city wise maximum salary.

```
SELECT City, MAX(Salary) AS MaxSalary  
FROM Employees  
GROUP BY City;
```

16. Produce output like <Ename> works at <city> and earns <salary>.

```
SELECT EName + ' works at ' + City + ' and earns ' +  
CAST(Salary AS VARCHAR(20)) AS Result  
FROM Employees;
```

17. Find total number of employees whose salary is more than 5000.

```
SELECT COUNT(*) AS TotalEmployees  
FROM Employees  
WHERE Salary > 5000;
```

18. Write a query to display first 4 & last 3 characters of all the employees.

```
SELECT LEFT(EName,4) + RIGHT(EName,3) AS Result  
FROM Employees;
```

19. List the city having total salaries more than 15000 and employees joined after 1st January, 2014.

```
SELECT City  
FROM Employees
```

```
WHERE JoiningDate > '2014-01-01'
GROUP BY City
HAVING SUM(Salary) > 15000;
```

20. Write a query to replace "u" with "oo" in Ename.

```
SELECT REPLACE(EName, 'u', 'oo') AS ModifiedName
FROM Employees;
```

21. Display city with average salaries and total number of employees belongs to each city.

```
SELECT City,
        AVG(Salary) AS AvgSalary,
        COUNT(*) AS TotalEmployees
FROM Employees
GROUP BY City;
```

22. Display total salaries paid to female employees.

```
SELECT SUM(Salary) AS TotalFemaleSalary
FROM Employees
WHERE Gender = 'Female';
```

23. Display name of the employees and their experience in years.

```
SELECT EName,
        DATEDIFF(YEAR, JoiningDate, GETDATE()) AS ExperienceYears
FROM Employees;
```

24. Remove column department from employees table.

```
ALTER TABLE Employees
DROP COLUMN Department;
```

25. Update the value of city from sydney to null.

```
UPDATE Employees
SET City = NULL
WHERE City = 'sydney';
```

26. Retrieve all the Employee's Name, Salary & Joining date.

```
SELECT EName, Salary, JoiningDate
FROM Employees;
```

27. Find out combine length of Ename & Gender.

```
SELECT LEN(EName) + LEN(Gender) AS CombinedLength
FROM Employees;
```

28. Find the difference between highest & lowest salary.

```
SELECT MAX(Salary) - MIN(Salary) AS SalaryDifference
FROM Employees;
```

29. Rename a column from Ename to FirstName.

```
EXEC sp_rename 'Employees.EName', 'FirstName', 'COLUMN';
```

30. Rename a table from Employees to EmpMaster.

`EXEC sp_rename 'Employees', 'EmpMaster';`